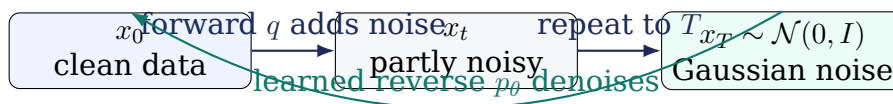


Chapter 3 Exam Study Guide

Diffusion Models

Forward Process, Reverse Process, DDPM, DDIM, Time Conditioning, U-Net



Purpose of this PDF

This guide summarizes the important exam topics from Chapter 3 using clear explanations, symbolic equations, sample exam answers, and flashcard-style revision questions.

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1 What you must know for the exam

Chapter 3 in one sentence

A diffusion model generates data by starting from simple Gaussian noise and learning a reverse denoising process that gradually moves $x_T \sim \mathcal{N}(0, I)$ back toward a clean sample $x_0 \sim q_{\text{data}}(x_0)$.

1.1 Learning objectives

By the end of this chapter, you should be able to explain:

1. Markov chains and trajectory notation.
2. The forward noising process q .
3. The closed-form marginal $q(x_t | x_0)$.
4. Why the true reverse process is hard.
5. How $p_\theta(x_{t-1} | x_t)$ is learned.
6. Why DDPM predicts noise ϵ .
7. Why time conditioning is required.
8. Why U-Net is useful for image diffusion.
9. The role of α_t , β_t , and $\bar{\alpha}_t$.
10. The difference between DDPM and DDIM.

2 Markov chain review

2.1 Markov property

A Markov chain is a sequence of random variables

$$x_0, x_1, x_2, \dots, x_T$$

that satisfies

$$p(x_t | x_{t-1}, x_{t-2}, \dots, x_0) = p(x_t | x_{t-1}).$$

In words, the next state depends only on the current state, not the full past.

2.2 Joint factorization

A Markov chain is specified by an initial distribution and a transition distribution:

$$p(x_0), \quad p(x_t | x_{t-1}).$$

The joint distribution factorizes as

$$p(x_{0:T}) = p(x_0) \prod_{t=1}^T p(x_t | x_{t-1}).$$

Exam intuition

This factorization is why Markov chains are tractable. Instead of modeling one huge joint distribution directly, we model many local conditional steps.

3 Core idea of diffusion models

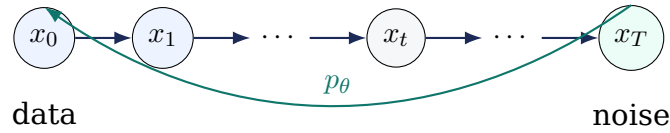
3.1 Forward and reverse directions

Diffusion uses two directions:

$$\text{Forward: } x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_T,$$

$$\text{Reverse: } x_T \rightarrow x_{T-1} \rightarrow \dots \rightarrow x_0.$$

The forward process gradually corrupts data into noise. The reverse process learns to denoise.



3.2 Why many steps?

Directly learning

$$z \sim \mathcal{N}(0, I) \longrightarrow x_0 \sim q_{\text{data}}$$

is difficult. Diffusion breaks generation into many local denoising steps. Each step only removes a small amount of noise, so the learning problem becomes more stable.

4 Notation: trajectory versus marginal

4.1 Trajectory notation

The notation

$$x_{1:T} = (x_1, x_2, \dots, x_T)$$

means the whole sequence of noisy states. It does not mean exponentiation or multiplication.

4.2 Trajectory distribution

The forward trajectory distribution is

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^T q(x_t \mid x_{t-1}).$$

This describes the probability of a full noisy path from x_0 to x_T .

4.3 Marginal distribution

The marginal distribution

$$q(x_t \mid x_0)$$

describes only the noisy state at time t , after integrating out the intermediate states:

$$q(x_t \mid x_0) = \int q(x_{1:t} \mid x_0) dx_1 \cdots dx_{t-1}.$$

Important distinction

Training usually uses the marginal $q(x_t \mid x_0)$ because it lets us jump directly from x_0 to any noisy timestep x_t . Sampling uses the reverse trajectory step by step.

5 Forward diffusion process in DDPM

5.1 One-step transition

In DDPM, the forward transition is Gaussian:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I).$$

Equivalently, by reparameterizing the Gaussian:

$$x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I).$$

5.2 Role of α_t and β_t

Usually,

$$\beta_t = 1 - \alpha_t, \quad 0 < \beta_t < 1.$$

So the same transition can be written as

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I),$$

$$x_t = \sqrt{1 - \beta_t}x_{t-1} + \sqrt{\beta_t}\epsilon_t.$$

Meaning of the parameters

α_t controls how much signal remains. β_t controls how much fresh noise is injected. A large β_t destroys information quickly; a very small β_t makes the process slow.

6 Closed-form marginal: the most important equation

6.1 Definition of $\bar{\alpha}_t$

Define the cumulative product

$$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s.$$

6.2 Direct sampling from x_0 to x_t

Because the forward process is linear Gaussian, we do not need to simulate every intermediate step. We can sample directly:

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I),$$

which implies

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

Why this matters

This closed-form marginal is what makes DDPM training practical. During training, we can randomly choose t , draw ϵ , construct x_t directly, and train the network without simulating $x_1, \dots, x_{t-1}$.

7 Reverse process and why it must be learned

7.1 True reverse transition

The ideal reverse transition is

$$q(x_{t-1} \mid x_t).$$

If we could sample this exactly from $x_T \sim \mathcal{N}(0, I)$ down to x_0 , we could generate data. The problem is that this conditional depends on the unknown data distribution $q_{\text{data}}(x_0)$, so it is not directly available for real datasets.

7.2 Learned reverse model

Diffusion models replace the true reverse transition with a neural approximation:

$$p_\theta(x_{t-1} \mid x_t) \approx q(x_{t-1} \mid x_t).$$

The full learned reverse process is

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1} \mid x_t), \quad p(x_T) = \mathcal{N}(0, I).$$

Often,

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)).$$

8 What the neural network predicts

8.1 Noise prediction target

DDPM commonly trains a network $\epsilon_\theta(x_t, t)$ to predict the noise ϵ used in

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon.$$

The simplified DDPM loss is

$$L_{\text{simple}}(\theta) = \mathbb{E}_{x_0, \epsilon, t} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2 \right], \quad \epsilon \sim \mathcal{N}(0, I).$$

Why noise prediction makes sense

If the model can estimate the noise component ϵ , then it knows the direction needed to move x_t back toward the clean data structure. For Gaussian corruption, noise prediction is closely related to score estimation.

8.2 Recovering an estimate of x_0

From the forward equation, if $\epsilon_\theta(x_t, t)$ is known, then

$$\hat{x}_0(x_t, t) = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}.$$

This expression is important in DDIM and in many sampling formulas.

8.3 Score connection

For

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I),$$

the conditional score is

$$\nabla_{x_t} \log q(x_t | x_0) = -\frac{x_t - \sqrt{\bar{\alpha}_t}x_0}{1 - \bar{\alpha}_t} = -\frac{\epsilon}{\sqrt{1 - \bar{\alpha}_t}}.$$

So predicting ϵ is equivalent to learning a scaled score direction under the Gaussian corruption model.

9 Timestep conditioning

9.1 Why t is necessary

The denoising task is not the same at every timestep. When t is small, x_t still looks close to x_0 . When t is large, x_t may be almost pure noise. Therefore the network must receive both x_t and t :

$$\epsilon_{\theta}(x_t, t).$$

Without t , the same noisy-looking input could require different denoising behavior depending on the signal-to-noise level.

9.2 Sinusoidal time encoding

A common time embedding is

$$\text{PE}(t, 2i) = \sin\left(\frac{t}{10000^{2i/d}}\right),$$

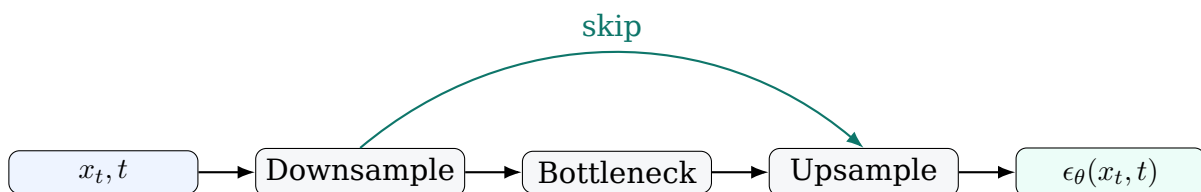
$$\text{PE}(t, 2i + 1) = \cos\left(\frac{t}{10000^{2i/d}}\right).$$

The embedding converts the scalar timestep t into a vector that the neural network can use.

10 Why U-Net is used

For images, denoising requires both global structure and local detail. A U-Net is suitable because it combines:

- a downsampling path that captures global context,
- an upsampling path that reconstructs fine spatial detail,
- skip connections that preserve high-resolution information.



11 Noise schedule

The noise schedule is the sequence

$$\{\beta_t\}_{t=1}^T \quad \text{or equivalently} \quad \{\alpha_t\}_{t=1}^T.$$

It controls how fast signal is destroyed. The cumulative signal strength is $\bar{\alpha}_t$ and the noise strength is $1 - \bar{\alpha}_t$.

11.1 Signal-to-noise ratio

A useful quantity is

$$\text{SNR}(t) = \frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t}.$$

At early timesteps, $\text{SNR}(t)$ is high. At late timesteps, $\text{SNR}(t)$ is low.

11.2 Linear versus cosine schedules

- A linear schedule increases β_t roughly linearly.
- A cosine schedule changes $\bar{\alpha}_t$ more smoothly and often preserves useful signal longer.

The schedule matters because it determines which timesteps carry semantic information and which timesteps are almost pure noise.

12 DDPM versus DDIM

12.1 DDPM

DDPM sampling is usually stochastic:

$$x_T \sim \mathcal{N}(0, I), \quad x_{t-1} \sim p_\theta(x_{t-1} | x_t).$$

At each reverse step, extra random noise may be injected.

12.2 DDIM

DDIM keeps the same training objective but changes sampling. It uses

$$\hat{x}_0(x_t, t) = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}},$$

and a reverse update of the form

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \hat{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \epsilon_\theta(x_t, t) + \sigma_t z, \quad z \sim \mathcal{N}(0, I).$$

The parameter η controls σ_t :

$$\eta = 0 \Rightarrow \text{deterministic DDIM}, \quad \eta = 1 \Rightarrow \text{more stochastic sampling}.$$

DDPM vs DDIM exam phrase

DDPM and DDIM can use the same trained noise predictor. The difference is mainly in sampling: DDPM is usually stochastic and slower; DDIM can be deterministic and faster with fewer steps.

13 Training and sampling algorithms

13.1 Training loop

1. Sample a clean data point $x_0 \sim q_{\text{data}}(x_0)$.
2. Sample a timestep $t \sim \text{Uniform}\{1, \dots, T\}$.
3. Sample noise $\epsilon \sim \mathcal{N}(0, I)$.
4. Create a noisy sample:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon.$$

5. Train $\epsilon_\theta(x_t, t)$ using

$$\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2.$$

13.2 Sampling loop

1. Start with $x_T \sim \mathcal{N}(0, I)$.
2. For $t = T, T-1, \dots, 1$, use $\epsilon_\theta(x_t, t)$ to denoise one step.
3. The final output x_0 is the generated sample.

14 Families of diffusion models

Family	Forward corruption	What is learned or changed
DDPM	Discrete Gaussian Markov chain	$\epsilon_\theta(x_t, t)$ or $\mu_\theta(x_t, t)$
DDIM	Same training as DDPM	Alternative sampler; can be deterministic
Score-SDE	Continuous-time SDE	Score $s_\theta(x, t) \approx \nabla_x \log p_t(x)$
Latent Diffusion	Diffusion in latent space z_t	Denoiser $\epsilon_\theta(z_t, t, c)$
Guided diffusion	Same forward process	Adds guidance during sampling
Consistency / distillation	Often learned from diffusion teacher	Direct or fewer-step mapping
Discrete diffusion	Masking or categorical corruption	Logits or denoising distribution

15 Formula sheet

Core equations to memorize

Markov property:	$p(x_t x_{t-1}, \dots, x_0) = p(x_t x_{t-1})$
Forward trajectory:	$q(x_{1:T} x_0) = \prod_{t=1}^T q(x_t x_{t-1})$
DDPM forward step:	$q(x_t x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I)$
Sampling step:	$x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\epsilon_t$
Noise schedule:	$\beta_t = 1 - \alpha_t, \quad \bar{\alpha}_t = \prod_{s=1}^t \alpha_s$
Closed-form marginal:	$q(x_t x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)$
Direct noising:	$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$
Reverse model:	$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1} x_t)$
Noise prediction loss:	$L_{\text{simple}} = \mathbb{E}_{x_0, \epsilon, t} [\ \epsilon - \epsilon_\theta(x_t, t)\ _2^2]$
Estimate clean sample:	$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}$
SNR:	$\text{SNR}(t) = \frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t}$

16 Sample exam questions with model answers

Question 1

What is the conceptual difference between $q(x_{1:T} | x_0)$ and $q(x_t | x_0)$, and why is that distinction important?

Model answer. $q(x_{1:T} | x_0)$ is the distribution over the entire noisy trajectory starting from x_0 . It describes the whole path $(x_1, \dots, x_T)$. By contrast, $q(x_t | x_0)$ is only the marginal distribution of one noisy state at timestep t . This matters because training usually only needs one noisy sample x_t , and the closed-form marginal lets us sample x_t directly without simulating all previous states.

Question 2

Why is the true reverse conditional $q(x_{t-1} | x_t)$ not directly usable in realistic diffusion modeling, even though the forward process is known?

Model answer. The forward conditional $q(x_t | x_{t-1})$ is known because we design it. However, the reverse conditional depends on the data distribution that produced the noisy sample. For complex data such as images, this distribution is unknown and intractable. Therefore, diffusion models learn a neural approximation $p_\theta(x_{t-1} | x_t)$.

Question 3

Why does predicting the added noise ϵ make sense as a training target?

Model answer. From the closed-form marginal,

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon.$$

Given x_t and t , the unknown corruption component is ϵ . Estimating ϵ tells the model how the clean sample was moved away from the data manifold. For Gaussian corruption, ϵ is also proportional to a score direction, so noise prediction is mathematically connected to denoising and reverse-time generation.

Question 4

Why must the denoising network know the timestep t ? Could one use only x_t ?

Model answer. Using only x_t is generally ambiguous because the correct denoising operation depends on the noise level. A small t means x_t still contains strong signal and requires gentle denoising. A large t means x_t is close to pure noise and requires a more global denoising step. Conditioning on t tells the model the current signal-to-noise regime.

Question 5

Why can diffusion models be described as both latent-variable models and score-based models?

Model answer. They are latent-variable models because the intermediate states $x_1, \dots, x_T$ form hidden variables in a long trajectory that mediates generation of x_0 . They are also score-based because the learned denoising direction can be interpreted as estimating gradients of log densities at noisy intermediate distributions. These views are compatible: one emphasizes the probabilistic trajectory, and the other emphasizes the geometric direction that moves samples toward high-density data regions.

17 Complete de-duplicated flashcard review

How this section was prepared

This section keeps the unique questions from the uploaded Chapter 3 flashcard deck and rewrites the answers with clean symbolic notation. Repeated ideas are kept only when the question tests a different exam angle.

No.	Flashcard question	Clean answer
1	What fundamental property defines a sequence of random variables as a Markov chain?	The future state depends only on the present state, not the past: $p(x_t x_{t-1}, \dots, x_0) = p(x_t x_{t-1})$.
2	Which two distributions are required to fully specify a Markov chain?	An initial distribution $p(x_0)$ and a transition distribution $p(x_t x_{t-1})$.
3	How does the joint distribution $p(x_{0:T})$ of a Markov chain factorize?	$p(x_{0:T}) = p(x_0) \prod_{t=1}^T p(x_t x_{t-1})$.
4	Under what condition does a reverse Markov chain typically exist if the forward process is Markov?	Under mild conditions, a reverse Markov chain exists for a Markov forward process.
5	What is the primary goal of a diffusion generative model?	To generate data samples by starting from a simple distribution, usually $x_T \sim \mathcal{N}(0, I)$, and reversing the noise process.
6	Which direction of the diffusion process is constructed by progressively adding small amounts of Gaussian noise?	The forward process: $x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_T$; data $\rightarrow$ Gaussian noise.
7	Why is the forward diffusion process easier to define than the reverse process?	It uses fixed, non-learned Gaussian transitions to simulate data corruption.

No.	Flashcard question	Clean answer
8	How does breaking generation into many small diffusion steps benefit the model?	Each step only removes a small amount of noise, making the learning problem more stable and well-conditioned.
9	Term: Trajectory.	The full sequence of intermediate noisy states: $x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_T$.
10	Notation: $x_{1:T}$.	$x_{1:T} = (x_1, x_2, \dots, x_T)$; the whole sequence from timestep 1 to T .
11	What does the marginal distribution $q(x_t x_0)$ represent in diffusion notation?	The probability of being at noisy state x_t after t noise steps, after integrating out all intermediate paths.
12	How is the DDPM forward transition $q(x_t x_{t-1})$ mathematically defined?	$q(x_t x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I)$.
13	What are the two effects of each DDPM forward diffusion step on the signal?	It shrinks the previous signal toward zero by $\sqrt{\alpha_t}$ and injects fresh Gaussian noise.
14	Using the reparameterization trick, how is a sample x_t generated from x_{t-1} ?	$x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\epsilon_t$, where $\epsilon_t \sim \mathcal{N}(0, I)$.
15	What formula defines the cumulative signal strength $\bar{\alpha}_t$ at timestep t ?	$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.
16	What is the closed-form Gaussian distribution for the marginal $q(x_t x_0)$?	$q(x_t x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)$.
17	As t approaches T in the forward process, what distribution does $q(x_t x_0)$ approach?	It approaches a standard normal distribution: $q(x_T x_0) \approx \mathcal{N}(0, I)$.
18	In the context of noise schedules, what does the parameter β_t represent?	β_t is the variance of the Gaussian noise injected at step t .
19	How is α_t related to the noise variance β_t in DDPM?	$\alpha_t = 1 - \beta_t$, equivalently $\beta_t = 1 - \alpha_t$.
20	What indicates that information is being destroyed during the forward diffusion process?	The mutual information $I(x_0; x_t)$ decreases toward zero as t increases.
21	How is the signal-to-noise ratio (SNR) at time t calculated in a diffusion model?	$\text{SNR}(t) = \frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t}$.
22	Which noise schedule provides smoother signal-to-noise decay compared with the linear schedule?	The cosine schedule.
23	Why is the true reverse transition $q(x_{t-1} x_t)$ considered intractable?	It depends on the unknown and complex data distribution $q_{\text{data}}(x_0)$.
24	By Bayes' rule, what components form the true reverse transition $q(x_{t-1} x_t)$?	$q(x_{t-1} x_t) = \frac{q(x_t x_{t-1})q(x_{t-1})}{q(x_t)}$; it uses the forward transition and a ratio of marginals.
25	What tractable distribution is often used in derivations as a target for the learned reverse model?	The Gaussian posterior $q(x_{t-1} x_t, x_0)$.
26	How is each learned reverse conditional $p_\theta(x_{t-1} x_t)$ usually parameterized?	As a Gaussian: $p_\theta(x_{t-1} x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$.
27	What are the two primary inputs provided to the neural network during the reverse process?	The current noisy sample x_t and the timestep t .
28	In the common DDPM parameterization, what does the neural network $\epsilon_\theta(x_t, t)$ actually predict?	It predicts the specific Gaussian noise ϵ that was added to x_0 to produce x_t .
29	How can an estimate of the clean data x_0 be recovered from the predicted noise $\epsilon_\theta(x_t, t)$?	$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}$.

No.	Flashcard question	Clean answer
30	What is the simplified DDPM training objective $L_{\text{simple}}(\theta)$?	$L_{\text{simple}}(\theta) = \mathbb{E}_{x_0, \epsilon, t} [\ \epsilon - \epsilon_\theta(x_t, t)\ _2^2]$.
31	Why is the U-Net an appropriate architecture for image-based diffusion processes?	It keeps input/output image resolution compatible and uses skip connections to preserve fine spatial details during denoising.
32	What is the purpose of skip connections in a diffusion U-Net?	They concatenate encoder features with decoder features to retain high-frequency details that may be lost during downsampling.
33	Why must diffusion models be conditioned on the timestep t ?	Denoising at small t is different from denoising at large t ; t tells the network the current noise level.
34	How are integer timesteps t typically represented before being fed into the neural network?	They are mapped to continuous vectors using sinusoidal time embeddings.
35	In a sinusoidal time embedding, what is the role of low-frequency dimensions?	They vary slowly with t and capture coarse temporal information.
36	Formula: SiLU (Swish) activation.	$\text{SiLU}(x) = x \sigma(x)$, where $\sigma(x) = \frac{1}{1 + e^{-x}}$.
37	How does the SiLU activation behave for large positive values of x ?	It behaves approximately like a linear function: $\text{SiLU}(x) \approx x$.
38	The standard linear noise schedule in the original DDPM typically ranges between which two β values?	Usually from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$.
39	In a ResBlock, how is the time embedding usually injected into the feature maps?	It is projected to match the channel count and then added as a per-channel bias.
40	What does the local-dependence property of a Markov chain imply graphically?	The chain is a sequence of nodes where each x_t has an incoming edge only from x_{t-1} .
41	How does forward diffusion affect the entropy of the data distribution?	It increases entropy as the data distribution becomes smoother and moves toward a simple Gaussian.
42	Term: Denoising Diffusion Probabilistic Models (DDPM).	A family of diffusion models using a Gaussian Markov chain for forward corruption and learned Gaussian transitions for reversal.
43	What is the starting point for generation during reverse sampling?	A sample $x_T \sim \mathcal{N}(0, I)$.
44	In the 1D mixture example, how does the distribution of x_t change as t increases from 0 to 199?	The bimodal two-bump distribution smooths out and becomes a single Gaussian centered at zero.
45	What is the relationship between $\bar{\alpha}_t$ and the total amount of noise accumulated in x_t ?	$1 - \bar{\alpha}_t$ is the variance of the total accumulated noise in x_t .
46	Why are sinusoidal embeddings preferred over directly feeding the integer t ?	They provide smooth interpolation across time and help the network generalize across timesteps.
47	In the reverse process, what happens if $t = 0$ is reached?	The model returns the final predicted mean as the generated clean sample x_0 .
48	What family of models diffuse in a latent space rather than directly on pixels?	Latent Diffusion Models (LDM).
49	Term: Score-SDE.	A continuous-time family of diffusion models that learns the score, i.e., the gradient of the log-density.

No.	Flashcard question	Clean answer
50	How is the marginal probability $q(x_t x_0)$ calculated by applying the law of marginalization?	$q(x_t x_0) = \int q(x_{1:t} x_0) dx_1 \cdots dx_{t-1}$.
51	What role does the bottleneck play in a U-Net architecture?	It captures the most global context by processing the image at its lowest spatial resolution.
52	In DDPM training, timesteps t are typically sampled from which distribution?	A discrete uniform distribution: $t \sim \text{Uniform}\{1, \dots, T\}$.
53	What is the primary difference between VE (Variance Exploding) and VP (Variance Preserving) schedules?	VP schedules keep the noisy-signal variance bounded, while VE schedules allow the variance to grow.
54	Which activation function is described as a smooth, non-monotonic gating of x ?	SiLU, also called Swish.
55	Why is it important that the forward process is a linear Gaussian transition?	It allows the marginal $q(x_t x_0)$ to be computed in closed form.
56	How does the law of marginalization justify the integral over paths in diffusion?	The marginal of x_t is obtained by integrating the joint trajectory distribution over all other intermediate variables.
57	In the 1D DDPM example, what does the tiny MLP output?	A single scalar representing the predicted noise added to x_t .
58	What happens to the SNR as the cumulative signal strength $\bar{\alpha}_t$ approaches zero?	$\text{SNR}(t) \rightarrow 0$, meaning the original signal is almost completely destroyed by noise.
59	Why is the reverse step $p_\theta(x_{t-1} x_t)$ stochastic rather than deterministic?	Many plausible clean signals could have produced the same noisy point x_t , so the model samples from a distribution.
60	Term: Consistency Models.	Diffusion-adjacent models, often trained from a teacher, that learn a direct or very-short-step mapping from noise to data.

18 Final exam revision checklist

Before the exam, make sure you can do the following without looking:

1. Write $q(x_t | x_{t-1})$ and explain each term.
2. Derive or explain $q(x_t | x_0)$ using $\bar{\alpha}_t$.
3. Explain why $q(x_{t-1} | x_t)$ is hard and why p_θ is needed.
4. Write the DDPM loss L_{simple} .
5. Explain why predicting ϵ is connected to denoising.
6. Explain why the network input is (x_t, t) , not only x_t .
7. Compare DDPM and DDIM in one paragraph.
8. Explain why U-Net fits image denoising.

Sources used. Chapter 3 - Diffusion Models lecture PDF; Chapter 3 - Summary and Sample Exam Questions PDF; Chapter 3 - Flashcards PDF.